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Saveetha Institute of Medical And Technical Sciences
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“ATHLETE-SHIELD AI: INTELLIGENT ATHLETE INJURY RISK PREDICTION & PERFORMANCE MONITORING”

Architecture Design Assignment

Submitted in the partial fulfilment for the Course of

CSA1012-SOFTWARE ENGINEERING

to the award of the degree of

BACHELOR OF TECHNOLOGY IN

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Submitted by

VENNAMPALLI BOOPATHY KHANISHK (192524311)

Under the Supervision of

Dr ANITA DAVAMANI K

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences Chennai-

602105

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INJURY RISK PREDICTION FOR ATHLETES USING PERFORMANCE DATA

1. INTRODUCTION

The Injury Risk Prediction System for Athletes Using Performance Data is designed to monitor athlete performance metrics and predict the likelihood of injuries before they occur. The system utilizes historical and real-time performance data, machine learning techniques, and analytical modules to identify injury risks and provide recommendations to coaches and athletes. The proposed solution helps improve athlete safety, optimize training schedules, and enhance overall performance.

2. SYSTEM OBJECTIVES

The primary objectives of the system are:

- To collect and manage athlete performance data.
- To analyze training and health-related metrics.
- To predict injury risks using machine learning algorithms.
- To provide risk scores and recommendations.
- To support coaches in making informed training decisions.
- To reduce injury occurrence and improve athlete well-being.

3. FUNCTIONAL REQUIREMENTS

➤ FR1 User Registration

Users shall be able to create accounts using valid credentials.

➤ FR2 User Authentication

The system shall provide secure login functionality.

➤ FR3 Athlete Profile Management

Users shall create and update athlete profiles.

➤ FR4 Performance Data Upload

The system shall allow athletes and coaches to upload performance data.

➤ FR5 Historical Data Storage

Historical athlete records shall be stored securely.

➤ FR6 Data Processing

The system shall clean, validate, and transform collected data.

➤ FR7 Injury Risk Prediction

The machine learning engine shall predict injury risks.

➤ FR8 Risk Score Generation

The system shall generate Low, Medium, or High risk scores.

➤ FR9 Alerts and Notifications

The system shall notify coaches and athletes about high-risk situations.

➤ FR10 Reports and Dashboard

The system shall generate reports and visual dashboards.

4. NON-FUNCTIONAL REQUIREMENTS

➤ Scalability

The system shall support a large number of athletes and coaches.

➤ Security

All athlete information shall be protected through authentication and secure storage.

➤ Performance

The prediction engine shall generate results within a few seconds.

➤ Reliability

The system shall provide consistent and accurate predictions.

➤ Availability

The application shall be accessible 24/7.

➤ Maintainability

The architecture shall allow easy updates and future enhancements.

5. QUALITY ATTRIBUTES ANALYSIS:

Quality Attribute	Description
Scalability	Supports growing numbers of users and records
Security	Protects sensitive athlete information
Performance	Provides fast prediction results
Reliability	Ensures dependable operation
Availability	Provides continuous service
Maintainability	Supports future upgrades

6. SELECTED SOFTWARE ARCHITECTURE

Hybrid Architecture (Layered Architecture + Machine Learning Service)

The proposed system adopts a Hybrid Architecture consisting of multiple layers and a dedicated machine learning prediction service.

The architecture includes:

- Presentation Layer
- Application Layer
- Data Processing Layer
- Injury Risk Prediction Engine
- Database Layer
- Alert and Reporting Layer

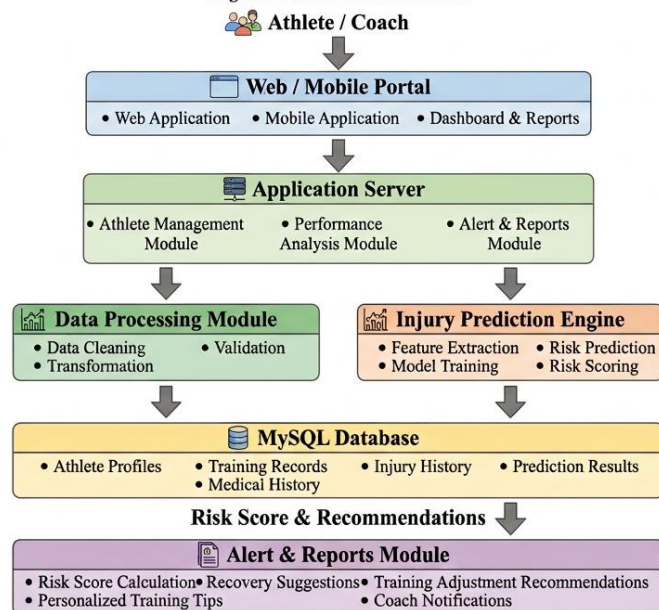
This architecture separates responsibilities and improves maintainability, scalability, and performance.

7. SOFTWARE ARCHITECTURE DIAGRAM

Figure 1: Architecture Diagram of Injury Risk Prediction System

Architecture Design for Injury Risk Prediction for Athletes Using Performance Data

Student Name: Vennampalli Boopathy Khanishk
Registration Number: 192524311



8. COMPONENT DESCRIPTION

Athlete / Coach

Provides input data and receives prediction results.

Web / Mobile Portal

Acts as the user interface for accessing the system.

Application Server

Handles business logic and communication between modules.

Data Processing Module

Performs data cleaning, validation, and transformation.

Injury Risk Prediction Engine

Executes machine learning algorithms and calculates risk scores.

MySQL Database

Stores athlete profiles, training records, medical history, and prediction results.

Alert and Reports Module

Generates recommendations and notifies coaches regarding potential risks.

9. COMPONENT INTERACTION WORKFLOW

1. Athlete uploads performance information.
2. Data is received through the web portal.
3. Application server processes requests.
4. Data Processing Module cleans and validates data.
5. Injury Risk Prediction Engine analyzes performance metrics.
6. Risk score is generated.
7. Results are stored in the MySQL database.
8. Alert and Reports Module generates recommendations.
9. Coaches and athletes receive notifications.

10. ARCHITECTURAL JUSTIFICATION

The Hybrid Architecture was selected because it effectively separates system functionalities into independent modules. The dedicated machine learning prediction engine enables advanced injury prediction without affecting other system components.

Advantages include:

- Better scalability
- Easier maintenance
- Improved reliability
- Modular development
- Future integration support

11. TRADE-OFF ANALYSIS

Benefits	Trade-Offs
High scalability	Slightly higher complexity
Modular design	More components to manage
Easier maintenance	Additional deployment effort
Better prediction capabilities	Requires machine learning resources

12. FUTURE ENHANCEMENTS

- Wearable sensor integration.
- Real-time monitoring system.
- Deep learning prediction models.
- Mobile application support.
- Cloud deployment.
- Advanced performance analytics.

13. CONCLUSION

The Injury Risk Prediction System for Athletes Using Performance Data provides a proactive approach to injury prevention by analyzing athlete performance metrics and predicting potential risks. The proposed Hybrid Architecture ensures scalability, maintainability, security, and reliability while supporting future enhancements and advanced analytics capabilities.

This report structure plus your diagram is exactly the type of content expected for the architecture design assignment rubric.